

Isometries 2

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Problem 1. *Why is this cool? Explain why it is surprising that a composition of rotations is a rotation – what data determines a rotation in $\mathbb{R}^3$?*

Proof. This is cool because it is not evident what would be axis of rotation. A rotation requires a axis of rotation and a angle so when you compose two rotations in $\mathbb{R}^n$ it is not evident what remains as the axis of rotation and what is the angle. $\square$

Problem 2. *Let A be a rotation, not the identity. Prove that $V_1(A)$ (the eigenspace) is one dimensional, the axis of rotation. If the angle is not π , show this is the only eigenspace, and that for π there are two eigenspaces, $V_1(A)$, which is one dimensional and $V_{-1}(A)$, which is 2 dim, orthogonal to $V_1(A)$.*

Proof. If A is a rotation and not the identity. By definition it has a axis of rotation around which the rotation occurs. The points of the axis are by definition fixed. Thus $L \in V_1(A)$. A rotation doesn't fix any other points thus $V_1(A) = L$.

If A is a rotation by $\theta \neq \pi$. If A cannot have any other eigenvalues because for all $v \notin V_1(A)$ v and Av are not collinear so 1 is the only real eigenvalue. If $\theta = \pm\pi$, since they are distinct then $\lambda = -1$ because points on the plane $\perp V_1(A)$ are also fixed, $V_1(A) \oplus V_{-1}(A) = \mathbb{R}^3$ thus $\dim V_{-1}(A) = 2$. $\square$

Problem 3. *Now let A be in $SO(3)$. We want to prove A is a rotation. The key is to find the 1-eigenspace. Prove that 1 is an eigenvalue (recall the complex eigenvalues have norm one and that eigenvalues are either real or come in conjugate pairs, and also the fact that a degree three real polynomial always has a real root). Let L be the span of a 1-eigenvector. Explain why P , the perp to L , is preserved by A , and why $A : P \rightarrow P$ is in $SO(2)$. Conclude $A : P \rightarrow P$ is a rotation, conclude A is a rotation. This proves Eulers theorem.*

Claim 3a. $\forall A \in SO(3)$, 1 is a eigenvalue of A .

Proof. Let $A \in SO(3)$. By definition, $\det A = 1$. Furthermore, characteristic polynomial $P_A(x)$ has degree 3 and real coefficients and since complex roots of $P_A(x)$ come in pairs or are real thus there is always a real root to $P_A(x)$. Thus A must have a real eigenvalue. Let $\lambda \in \mathbb{R}$ be a eigenvalue of A , $\det(A - I) = \det((A - I)^T) = \det(A^T - I) = \det(A^T - A^T A) = -\det(A^T) \det(A - I) = -\det(A - I)$. Thus $\det(A - I) = 0$. Thus 1 is a eigenvalue of A . $\square$

Claim 3b. *Let L be the span of a 1-eigenvector. Explain why P , the perp to L , is preserved by A , and why $A : P \rightarrow P$ is in $SO(2)$. Conclude $A : P \rightarrow P$ is a rotation, conclude A is a rotation. This proves Eulers theorem.*

Proof. Let L be the span of a 1-eigenvector, v . Let P be the subspace such that $L \perp P$. $\forall p \in P$, $\langle p, v \rangle = 0 = \langle Ap, Av \rangle$ since A preserves inner product. Thus $\forall p \in P$, $p \perp v$. Since v is preserved, the plane P is preserved as well. A can be made block diagonal s.t $A = \begin{bmatrix} 1 & 0 \\ 0 & B \end{bmatrix}$, corresponds to the L , then B corresponds to P . Since $\det A = 1$ then $\det B = 1$ and since A was orthogonal so is B thus $B \in SO(2)$ and is a rotation. Thus A must be a rotation as well. $\square$

Claim 4. For W in $O(3)$, show $W\rho_{v,\theta}W^{-1} = \rho_{Wv,\pm\theta}$ - meaning we either get θ or $-\theta$.

Proof. $W\rho_{v,\theta}W^{-1}$ must be a rotation by the determinant. Since $\det(W\rho_{v,\theta}W^{-1}) = 1$.

$W\rho_{v,\theta}W^{-1}(Wv) = W\rho_{v,\theta}v = Wv$. Thus Wv is fixed. The angle of a rotation is defined by its nonreal eigenvalue. All eigenvalues of $\rho_{v,\theta}$ are 1 and $e^{\pm i\theta}$ and eigenvalues don't change under conjugation. So $W\rho_{v,\theta}W^{-1} = \rho_{Wv,\pm\theta}$ $\square$

Claim 5. Use the fact that commuting matrices preserve each others eigenspaces to show that for non-identity A in $SO(3)$, the "centralizer" of A in $SO(3)$ (elements that commute with A) are exactly those rotations with the same axis, except when the angle for A is π . In this last case show that $SO(2)$ (the rotations with the same axis as A) is index 2 in $Z(A)$, the centralizer of A , and that $Z(A)$ is naturally isomorphic to $O(2)$.

Proof. $\forall A \in SO(3) \ni A \neq I, A \neq \rho_{v,\pi}. \forall v \in \mathbb{R}^3. \forall B \in SO(3) \ni AB = BA$. Since commuting matrices preserve each other's eigenspace then, if nonzero $v \in V_1(A)$, then $Bv \in V_1(A)$. Thus B preserves axis of rotation as A . So v is a eigenvector of B . So either B has the same axis as A or $B = \rho_{w,\pi}$ where $w \perp v$. If v is the same axis then $B \in SO(2)$. If $B = \rho_{w,\pi}$ since A and B commute, then A preserves the eigenspace of B , where $BAw = Aw$ thus $Aw \in V_1(B)$ but $\dim V_1(B) = 1$ so $Aw = \lambda w$ and since $\theta \neq \pm\pi$ since A only has 1 eigenvalue 1 with eigenvector v and $v \perp w$ thus a contradiction occurs. Thus B can be any rotation around the same axis thus all of $SO(2)$.

If $A = \rho_{v,\pi}$. Then the first case of the previous problem still applies, $SO(2) \in Z(A)$. However we also have $\rho_{w,\pi} \in Z(A)$. $\rho_{w,\pi}$ acting on the plane $P \perp v$ is the same thing as a reflection. So all rotations plus 1 reflections, generate $O(2)$ thus $Z(A) \cong O(2)$.

$\forall A \in SO(3) \ni A = \rho_v(v, \pi), v \neq 0$. In the plane subspace $P \perp L_v$ (line spanned by v). $A : P \rightarrow P$, $A = \rho_\pi$. $SO(2) \in Z(A)$. $\forall b \in Z(A) \ni B \notin SO(2)$, $bSO(2)$ are all reflections in $O(2)$. Thus $[Z(A) : SO(2)] = 2$. Thus $Z(A) \cong O(2)$. $\square$

Claim 6. Give a formula for conjugation in $Rig(\mathbb{R}^2)$, which Artin calls "M", in terms of his classification 6.3.4 — that is generalize our formula that $Ar_LA^{-1} = r_{A(L)}$ in $O(2)$ (where r_L is reflection about the line L).

Proof. By the previous homework, $\forall A \in Rig(\mathbb{R}^2)$, $A = t_v \circ B$ for $B \in O(2)$. $Ar_LA^{-1} = t_v \circ Br_LB^{-1} \circ t_{-v} = t_v \circ r_{B(L)} \circ t_{-v} = r_{B(L)+v}$ or reflection about line $B(L) + v$. $A\rho_\theta A^{-1} = t_v \circ B\rho_\theta B^{-1} \circ t_{-v} = t_v \circ \rho_{\pm\theta} \circ t_{-v}$ is rotation of θ about the point v . $\square$